



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 – 64



**KOLBI Cassava Chipper/Slicer
Model: KCSL24**

KOLBI KCSL24 CASSAVA CHIPPER/SLICER

Test Report No.	:	2013 - 64
Date of Test	:	May 11, 2013
Test Valid Until	:	May 11, 2018
Brand	:	KOLBI
Model	:	KCSL24
Type	:	Flywheel-type
Manufacturer	:	TROPICS AGRO INDUSTRIES, INC. 25 Panganiban Drive Naga City
Test Requested by	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The KOLBI KCSL24 Cassava Chipper/Slicer powered by a 7.40 kW MITSUBISHI GB300 gasoline engine was tested using freshly harvested cassava with an average Moisture Content of 57.4%.

The KOLBI KCSL24 Cassava Chipper/Slicer had an average Chipping Capacity of 1508 kg/h cassava tubers at chipper shaft speed of 294 rpm. The chips had an average thickness of 5.9 mm with thickness variation of $\pm 1.60\%$.

The noise emitted by the machine at 50 mm away from the operator's ear level was 91.2 db (A). The Fuel Consumption Rate of the 7.4 kW MITSUBISHI GB300 gasoline engine was 2.36 L/h.

Three persons were needed to operate the machine: one for hauling the cassava; one for feeding; and, one for collecting the cassava chips.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance. The test was conducted at the plant site of TROPICS AGRO-INDUSTRIES, INC. in Naga City on May 11, 2013. The test was carried out to assess and verify the following:

1. Machine specifications
2. Performance
 - a. Chipping capacity
 - b. Speed of components
 - c. Noise level
 - d. Fuel Consumption
3. Labor requirement
4. Quality of Work
 - a. Average thickness of chips
 - b. Variation in chips thickness

OPERATOR'S MANUAL

An Operation and Instruction Manual was submitted which contains the following: Application; Operation; Safety Instruction; Blade Adjustment, and, Trouble Shooting.

DESCRIPTION OF THE MACHINE

The KOLBI KCSL24 Cassava/Slicer as shown in **Figure 1**, was flywheel-type chipping unit. It was powered by a 7.4 kW MITSUBISHI GB300 gasoline engine which drove an intermediate shaft through a single groove B-section pulley and V-belt. The intermediate shaft in turn drove the chipper shaft through a single groove B-section pulley and V-belt. Belt power transmission system of the rice thresher is shown in **Figure 2**.

The machine consisted of the hopper and the chipping chamber. The hopper (**Fig. 3**) was where the materials to be chipped were fed. It was made of G. I. sheet, Ga. #18 and installed above the chipping chamber.

The chipping chamber consisted of a chipping plate (**Fig. 4a**) with three pieces of chipping blades (**Fig. 4b**). The chipping plate had dimensions (D x T) of 610 mm x 6 mm and made of M. S. plate, while the blades had dimensions (L x W x T) of 175 mm x 55 mm x 3 mm and made of stainless steel. Below the chipping chamber was the outlet chute (**Fig. 5**) where cassava chips flow down.

The whole frame of the machine was made of 51 mm x 51 mm x 6 mm angle bar.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. KOLBI Cassava Chipper/Slicer
Model: KCSL24

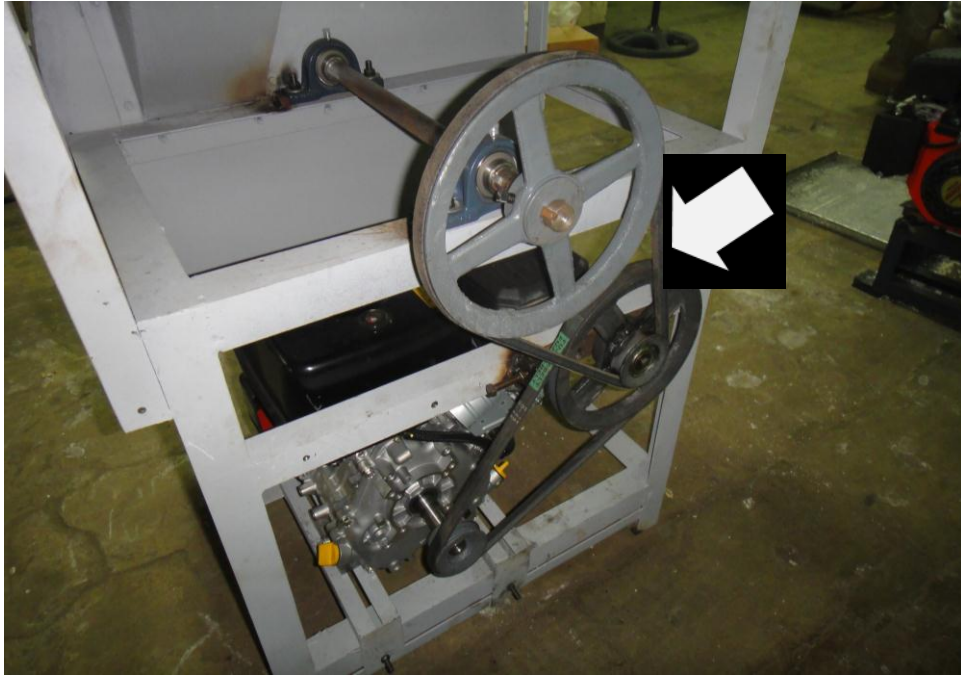


Figure 2. Belt-pulley transmission system

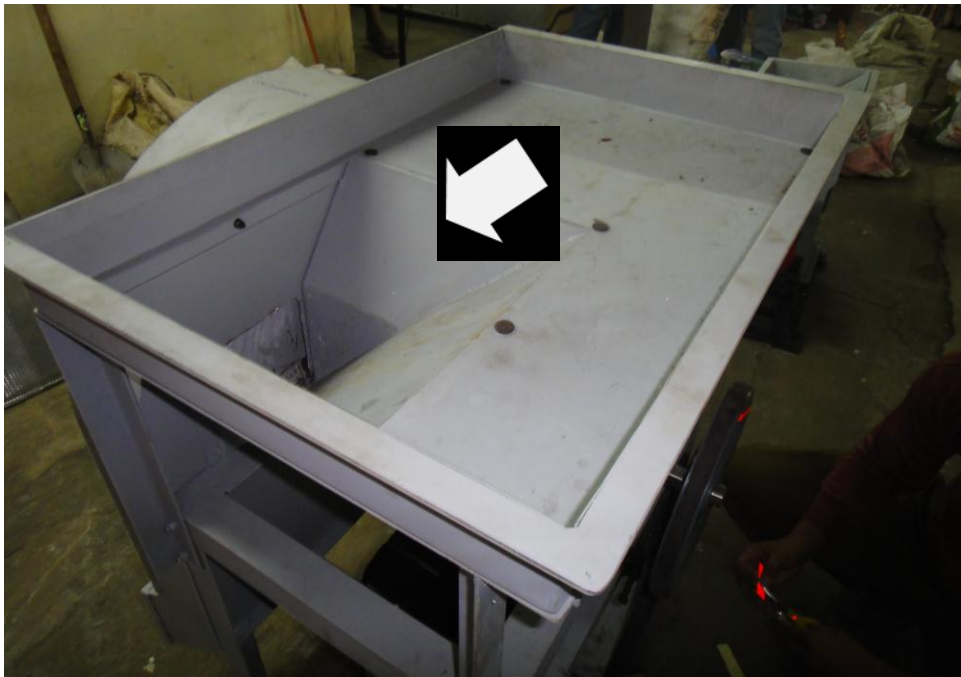


Figure 3. Hopper

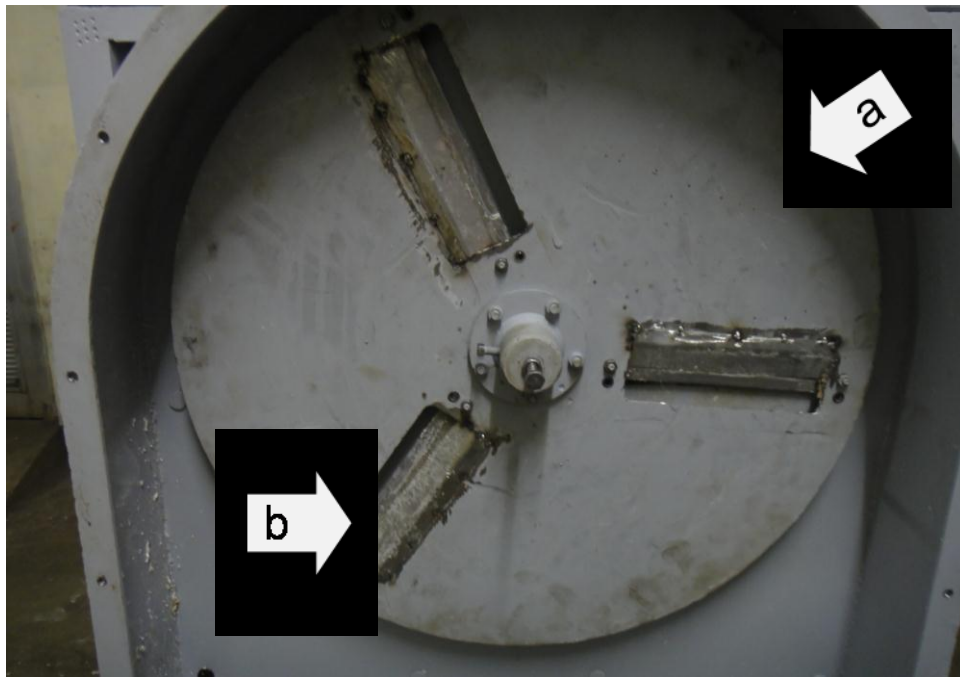


Figure 4. a) Chipper plate
b) Chipper blade



Figure 5. Outlet chute

SPECIFICATIONS

Table 1. Detailed specifications of the KOLBI KCSL24 Cassava Chipper/Slicer

A.	Brand	:	KOLBI
B.	Model	:	KCSL24
C.	Manufacturer	:	TROPICS AGRO INDUSTRIES, INC. 25 Panganiban Drive Naga City
D.	Overall Dimensions and Weight		
1.	Length (mm)	:	765
2.	Width (mm)	:	605
3.	Height (mm)	:	1100
4.	Weight (kg) without engine	:	No Data
E.	Hopper		
1.	Inlet opening, L x W (mm)	:	160 x 155
2.	Height from the ground (mm)	:	1100
3.	Material	:	G. I. Sheet, Ga. #18
F.	Chipper Plate		
1.	Size, D x T (mm)	:	610 x 6
2.	Material	:	M. S. Plate
G.	Chipper Blade		
1.	No. of pieces	:	3

Table 1. Detailed specifications of the KOLBI KCSL24 Cassava Chipper (cont'n.)

	2.	Size, L x W x T (mm)	:	175 x 55 x 3
	3.	Material	:	Stainless
H.		Discharge Chute		
	1.	Size, L x W (mm)	:	695 x 155
	2.	Height from the ground (mm)	:	215
	3.	Material	:	G. I. Sheet, Ga. #18
I.		Main Frame		
	1.	Material	:	Angle bar (51 mm x 51 mm x 6 mm)
J.		Engine (Prime mover)		
	1.	Brand	:	MITSUBISHI
	2.	Model	:	GB300
	3.	Make	:	Japan
	4.	Type	:	Four stroke, air-cooled single cylinder gasoline engine
	5.	Rated Maximum Power (kW)	:	7.4
	6.	Rated Speed (rpm)	:	3600
	7.	Displacement (cc)	:	296
	8.	Cooling system	:	Air-cooled
	9.	Starting system	:	Rope recoil
	10.	Dry weight (kg)	:	29.0

RESULTS AND DISCUSSIONS

Results of performance tests are shown in **Table 2-3**.

Chipping Capacity

It refers to the amount of input material that can be chipped per unit time. Test results showed that using freshly harvested cassava, the machine had a Chipping Capacity of 1508 kg/h cassava tubers at blade shaft speed of 294 rpm.

Quality of Work

The chips produced by the chipper had an average thickness of cut of 5.9 mm with Thickness Variation of ± 1.60 .

Noise Level

The noise emitted by the machine 50 mm away from the operator's ear level was 91.2 db(A).

Fuel Consumption

The 7.4 kW MITSUBISHI GB300 gasoline engine had a Fuel Consumption Rate of 2.36 L/h.

Labor Requirement

Three (**Fig. 6**) persons were needed to operate the machine: one for hauling the cassava; one for feeding; and, one for collecting the cassava chips.

Observations:

1. Freshly harvested cassava tubers (**Figure 7**) were used in the test. The chipper produced cassava chips with average thickness of 5.9 mm (**Figure 8**).
2. The chipper was provided with belt cover (**Figure 9**).



Figure 6. Operators of the machine



Figure 7. Cassava used in the test



Figure 8. Cassava chips produced by the chipper

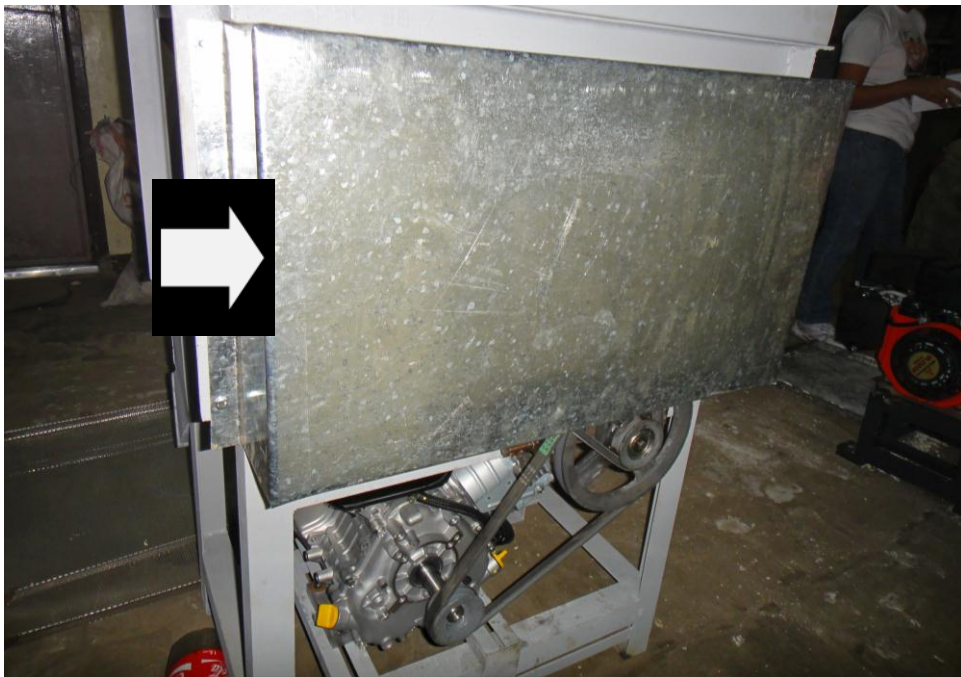


Figure 9. Belt cover

PERFORMANCE TEST

Machine Tested : KOLBI Cassava Chipper/Slicer
Model: KCSL24

Date of Test : May 11, 2013

Location of Test : Naga City, Camarines Sur

Table 2. Conditions of Test Materials

ITEMS	Data
1. Crop	Cassava
2. Variety	“Lakan”
3. Moisture Content, $MC_{w.b.}$ (%)	57.4
4. Dimension, L x D (mm)	315 x 48
5. Source	Tigaon, Camarines Sur

Table 3. Results of Performance Test

ITEMS	Data
1. Speed of Components (rpm)	
a) Engine (rpm)	
Without load	3016
With load	2992
b) Intermediate Shaft	
Without load	1025
With load	1010
c) Chipper Shaft	
Without load	301
With load	294

Table 3. Results of Performance Test (cont'n.)

ITEMS	Data
2. Noise Level, [db(A)]	
a) Operator	
Without load	87.6
With load	91.2
b) Bagger	
Without load	90.5
With load	93.0
3. Prime mover's Fuel Consumption (MITSUBISHI GB300), L/h	2.36
4. Average thickness of cut, mm	5.90
5. Variation in thickness, %	± 1.60
6. Chipping Capacity, kg/h	1508

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DARWIN C. ARANGUREN

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ROMULO E. EUSEBIO

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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